

Assignment 4

1. Implement the non-preemptive SJF scheduling algorithm.

Input:

A list of processes. Each process has:

- Process ID (e.g., P1, P2, P3)
- Arrival Time (when the process enters the ready queue)
- Burst Time (the total CPU time required to execute the process)

The list should be sorted based on arrival time

Example Input:

Process ID | Arrival Time | Burst Time

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P1	0	8
P2	1	4
P3	2	9
P4	3	5

Output:

A Gantt chart showing the execution order of processes.

Average Waiting Time

Average Turnaround Time

Gantt Chart:

P1 (0-8) | P2 (8-12) | P4 (12-17) | P3 (17-26)

Average Waiting Time: $(0 + 7 + 15 + 9) / 4 = 7.75$

Average Turnaround Time: $(8 + 11 + 24 + 14) / 4 = 14.25$

2. Implement the Round Robin scheduling algorithm.

Input:

List of processes with Process ID, Arrival Time, Burst Time.

Time Quantum (the amount of time each process gets to run before being preempted).

Output:

Gantt chart.

Average waiting time.

Average turnaround time.